

MINSI SUNG

Senior R&D Engineer with 6 Years of Experience in C++

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EXPERIENCE

Senior R&D Engineer

ECAD Team, Ansys

📅 Jun 2021 – Present

📍 Vancouver, Canada

- Develop and maintain large-scale commercial ECAD software, specializing in geometric operations and UI design using **C++** and **MFC** on Windows and Linux.
- Researched and integrated **Boost Polygon** for union operations to overcome computing bottlenecks in large IC layout design files. It achieved a 20x performance boost and got adopted across multiple products.
- Developed an algorithm to automatically resolve mesh region intersections to enhance user workflow and reduce simulation time.

Software Engineer Intern

Lumerical, Ansys

📅 Feb 2021 – May 2021

📍 Vancouver, Canada

- Implemented geometry methods using **Parasolid** and designed the user interface to interact and display the result using **C++** and **Qt** in **Linux** environment.

Research Assistant

CAD/CAM Lab, University of British Columbia

📅 Sept 2018 – Jan 2021

📍 Vancouver, Canada

- Published an automatic machine tool components grouping method using voxel-based collision detection that boosted efficiency by 20% and solved real-world challenges.
- Developed a 3D interactive environment to read STL files and the generated kinematic chains of machine tools for movements simulation in **C++** with **OpenGL** in **Qt**.

PROJECTS

Parallel Sorting Optimization Using CUDA & NSight Compute (Project of CS8803 GPU Hardware & Software)

📅 Jan 2025 – Apr 2025

- Implemented Bitonic Sort using **CUDA**, optimizing shared memory kernels and leveraging pinned memory for efficient data transfer.
- Analyzed performance bottlenecks using **NVIDIA Nsight Compute** to optimize **GPU** occupancy and memory throughput to achieve a 180x speedup over `std::sort` on CPU.

Multithreading, Inter-Process Communication

(Projects of CS 6200 Graduate Introduction to Operating Systems)

📅 Sep 2023 – Dec 2023

- Implemented boss/worker model multi-thread server and client using **pthread** in **C** to overcome the file transferring performance issue.
- Utilized **POSIX** share memory API for shared memory-based IPC to communicate and transfer files between proxy and cache for the requests from clients.

Remesher with Four Processes for Large Triangular Mesh Models (Final project of CPSC 524 Computer Graphics: Modelling)

📅 Feb 2019 – Jun 2019

- Completed the remesh process with a user friendly API in **C++** by iterating through 3D geometry models for mesh refinement, edge collapse, edge flipping and smoothing.

SKILLS

Programming Languages

C/C++, Python, C#, Javascript

Tools

CUDA, OpenGL, Qt, React, MFC, Git, Visual Studio

LEADERSHIP

President

Taiwanese Graduate Student Association in Vancouver

📅 Feb 2020 – Feb 2021

📍 Vancouver, Canada

Captain

Baseball Team in Mechanical Engineering Department

📅 July 2016 – July 2017

📍 Tainan, Taiwan

EDUCATION

Master of Science in Computer Science

Georgia Institute of Technology United States

📅 Aug 2023 – Expected Dec 2025

Master of Applied Science in Mechanical Engineering University of British Columbia Canada

📅 Sep 2018 – Jan 2021

Bachelor of Engineering in Mechanical Engineering National Cheng Kung University Taiwan

📅 Sep 2013 – Jan 2018